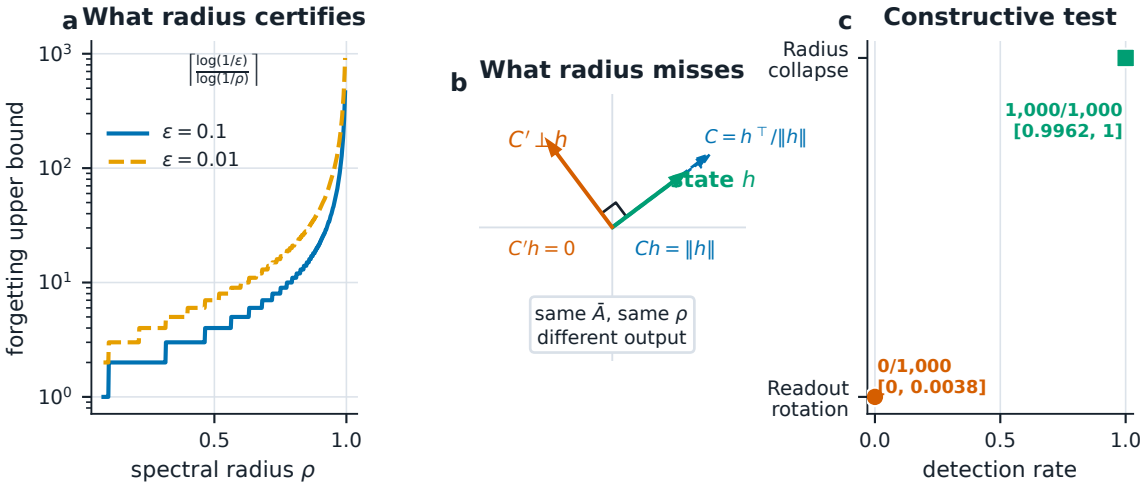
a
What radius certifies
$\left\lceil \frac{\log(1/\varepsilon)}{\log(1/\rho)} \right\rceil$
$\varepsilon = 0.1$
$\varepsilon = 0.01$
forgetting upper bound
10^3
10^2
10^1
10^0
0.5
1.0
spectral radius ρ
b
What radius misses
$C' \perp h$
$C = h^\top / \|h\|$
state h
$C'h = 0$
$Ch = \|h\|$
same $\bar{A}$, same ρ
different output
c
Constructive test
Radius collapse
1,000/1,000
[0.9962, 1]
0/1,000
[0, 0.0038]
Readout rotation
0.0
0.5
1.0
detection rate